

SHETH L.U.J AND SIR MV COLLEGE

PRACTICAL NO. 08

AIM :- Practical Objective: Set up and configure an open-source Intrusion Detection System to monitor network traffic and detect potential security breaches or malicious activities

Step 1: Update Ubuntu

sudo apt install snort -y

```
rani@LAPTOP-OMML3RLS:/n x + v
C:\Users\Rani>wsl --install -d Ubuntu-24.04
Downloading: Ubuntu 24.04 LTS
Installing: Ubuntu 24.04 LTS
Distribution successfully installed. It can be launched via 'wsl.exe -d Ubuntu-24.04'
Launching Ubuntu-24.04...
Provisioning the new WSL instance Ubuntu-24.04
This might take a while...
Create a default Unix user account: rani
New password:
Retype new password:
passwd: password updated successfully
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

rani@LAPTOP-OMML3RLS:/mnt/c/Users/Rani$ sudo apt update
[sudo] password for rani:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [971 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [206 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [46.4 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [11.9 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1203 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [240 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [76.3 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [24.2 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [1392 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [321 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:17 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 c-n-f Metadata [444 B]
Get:18 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [40.3 kB]
Get:19 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [11.1 kB]
Get:20 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [208 B]
Get:21 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [468 B]
Get:22 http://archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:23 http://archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:24 http://archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
```

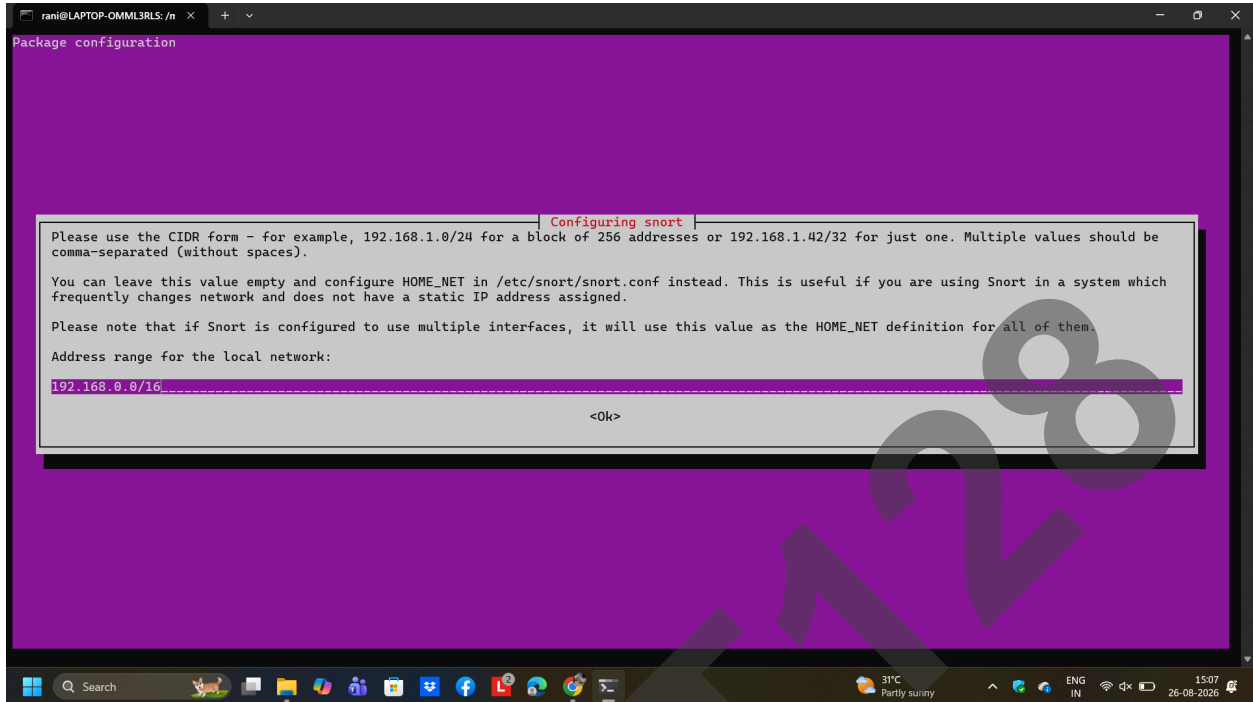
Step 2: Install Snort

sudo apt install snort -y

```
rani@LAPTOP-OMML3RLS:/n x + v
rani@LAPTOP-OMML3RLS:/mnt/c/Users/Rani$ sudo apt install snort -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ibverbs-providers libauthen-sasl-perl libclone-perl libdaq2t64 libdata-dump-perl libdumbnet1 libencode-locale-perl libfile-listing-perl libfont-afm-perl
  libhtml-form-perl libhtml-format-perl libhtml-parser-perl libhtml-tagset-perl libhtml-tree-perl libhttp-cookies-perl libhttp-daemon-perl
  libhttp-date-perl libhttp-message-perl libhttp-negotiate-perl libibverbs1 libio-html-perl libio-socket-ssl-perl liblua5.1-2 liblua5.1-common
  liblwp-mediatypes-perl liblwp-protocol-https-perl libmailtools-perl libnet-http-perl libnet-smtp-ssl-perl libnet-ssleay-perl libnetfilter-queue1
  libnftnl0 libnl-3-200 libnl-route-3-200 libpcap0.8t64 libpcre3 libperl5.38t64 libtimedate-perl libtiny-perl liburi-perl libwww-perl libwww-robotrules-perl
  net-tools oinkmaster perl perl-base perl-modules-5.38 perl-openssl-defaults snort-common snort-common-libraries snort-rules-default
Suggested packages:
  libdigest-hmac-perl libgssapi-perl libio-compress-brotli-perl libcrypt-ssleay-perl libsub-name-perl libbusiness-isbn-perl libregexp-ipv6-perl
  libauthen-ntlm-perl perl-doc libterm-readline-gnu-perl | libterm-readline-perl-perl make libtap-harness-archive-perl debhelper snort-doc
The following NEW packages will be installed:
  ibverbs-providers libauthen-sasl-perl libclone-perl libdaq2t64 libdata-dump-perl libdumbnet1 libencode-locale-perl libfile-listing-perl libfont-afm-perl
  libhtml-form-perl libhtml-format-perl libhtml-parser-perl libhtml-tagset-perl libhtml-tree-perl libhttp-cookies-perl libhttp-daemon-perl
  libhttp-date-perl libhttp-message-perl libhttp-negotiate-perl libibverbs1 libio-html-perl libio-socket-ssl-perl liblua5.1-2 liblua5.1-common
  liblwp-mediatypes-perl liblwp-protocol-https-perl libmailtools-perl libnet-http-perl libnet-smtp-ssl-perl libnet-ssleay-perl libnetfilter-queue1
  libnftnl0 libnl-3-200 libnl-route-3-200 libpcap0.8t64 libpcre3 libtimedate-perl libtiny-perl liburi-perl libwww-perl libwww-robotrules-perl
  net-tools oinkmaster perl perl-base perl-modules-5.38 perl-openssl-defaults snort-common snort-common-libraries snort-rules-default
The following packages will be upgraded:
  libperl5.38t64 perl perl-base perl-modules-5.38
4 upgraded, 48 newly installed, 0 to remove and 159 not upgraded.
Need to get 15.4 MB of archives.
After this operation, 20.0 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libperl5.38t64 amd64 5.38.2-3.2ubuntu0.3 [4876 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 perl amd64 5.38.2-3.2ubuntu0.3 [231 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 perl-base amd64 5.38.2-3.2ubuntu0.3 [1827 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 perl-modules-5.38 all 5.38.2-3.2ubuntu0.3 [3110 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 liblua5.1-2-common all 2.1.0+git20231223.c525bcb+dfsg-1ubuntu0.1 [49.5 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 liblua5.1-2 amd64 2.1.0+git20231223.c525bcb+dfsg-1ubuntu0.1 [276 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble/universe amd64 libpcre3 amd64 2:8.39-15build1 [248 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble/universe amd64 snort-common-libraries amd64 2.9.20-0+deb11u1ubuntu1 [899 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble/universe amd64 snort-rules-default all 2.9.20-0+deb11u1ubuntu1 [144 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/universe amd64 snort-common all 2.9.20-0+deb11u1ubuntu1 [47.7 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 net-tools amd64 2.10-0.1ubuntu4.4 [204 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble/universe amd64 libdumbnet1 amd64 1.17.0-1ubuntu2 [30.7 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble/main amd64 libnftnl0 amd64 1.0.2-2build1 [14.8 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble/universe amd64 libnetfilter-queue1 amd64 1.0.5-4build1 [15.1 kB]
```

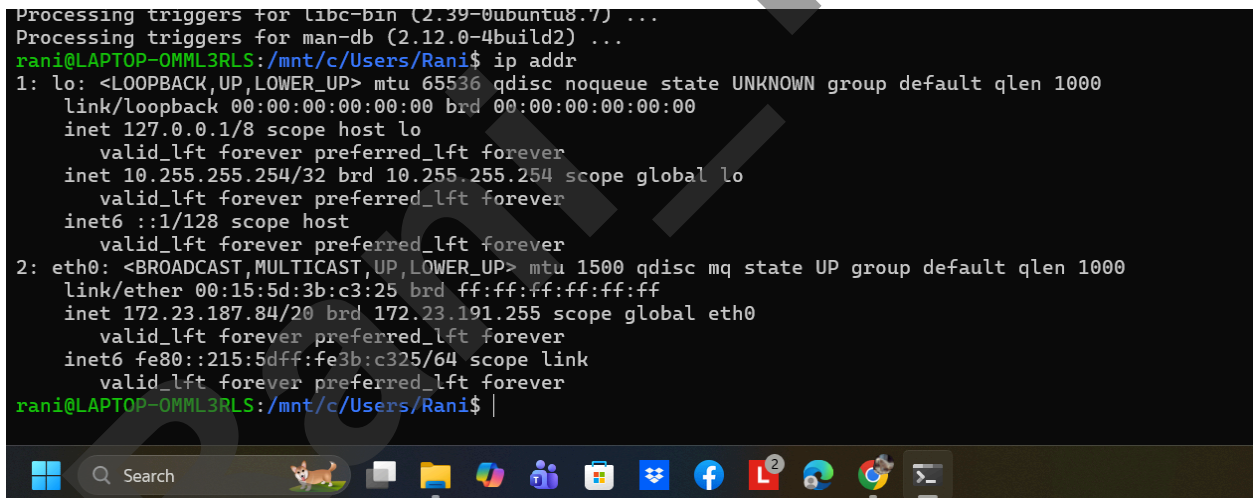
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Step 3: Check Network Interface

ip addr



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Step 4: Test Snort Configuration

sudo snort -T -c /etc/snort/snort.conf

```
rani@LAPTOP-OMML3RLS: /n  +  v
valid_lft forever preferred_lft forever
rani@LAPTOP-OMML3RLS: /mnt/c/Users/Rani$ sudo snort -T -c /etc/snort/snort.conf
Running in Test mode

--== Initializing Snort ==--
Initializing Output Plugins!
Initializing Preprocessors!
Initializing Plug-ins!
Parsing Rules file "/etc/snort/snort.conf"
PortVar 'HTTP_PORTS' defined : [ 80:81 311 383 591 593 901 1220 1414 1741 1830 2301 2381 2809 3037 3128 3702 4343 4848 5250 6988 7000:7001 7144:7145 7510 7
777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8888 8899 9000 9060 9080 9090:9091 9443 9999 11371 34443:34444 4108
0 50002 55555 ]
PortVar 'SHELLCODE_PORTS' defined : [ 0:79 81:65535 ]
PortVar 'ORACLE_PORTS' defined : [ 1024:65535 ]
PortVar 'SSH_PORTS' defined : [ 22 ]
PortVar 'FTP_PORTS' defined : [ 21 2100 3535 ]
PortVar 'SIP_PORTS' defined : [ 5060:5061 5600 ]
PortVar 'FILE_DATA_PORTS' defined : [ 80:81 110 143 311 383 591 593 901 1220 1414 1741 1830 2301 2381 2809 3037 3128 3702 4343 4848 5250 6988 7000:7001 714
4:7145 7510 7777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8888 8899 9000 9060 9080 9090:9091 9443 9999 11371 344
43:34444 41080 50002 55555 ]
PortVar 'GTP_PORTS' defined : [ 2123 2152 3386 ]
Detection:
  Search-Method = AC-Full-Q
  Split Any/Any group = enabled
  Search-Method-Optimizations = enabled
  Maximum pattern length = 20
  Tagged Packet Limit: 256
Loading dynamic engine /usr/lib/snort/snort_dynamicengine/libsfe_engine.so... done
Loading all dynamic detection libs from /usr/lib/snort/snort_dynamicrules...
WARNING: No dynamic libraries found in directory /usr/lib/snort/snort_dynamicrules.
  Finished Loading all dynamic detection libs from /usr/lib/snort/snort_dynamicrules.
Loading all dynamic preprocessor libs from /usr/lib/snort/snort_dynamicpreprocessor/...
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_appid_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_modbus_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_gtp_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_ftptelnet_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_ssh_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_sdf_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_dnp3_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_ssl_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_sntp_preproc.so... done
```

```
rani@LAPTOP-OMML3RLS: /n  +  v
SSH config:
Autodetection: ENABLED
Challenge-Response Overflow Alert: ENABLED
SSH1 CRC32 Alert: ENABLED
Server Version String Overflow Alert: ENABLED
Protocol Mismatch Alert: ENABLED
Bad Message Direction Alert: DISABLED
Bad Payload Size Alert: DISABLED
Unrecognized Version Alert: DISABLED
Max Encrypted Packets: 20
Max Server Version String Length: 100
MaxClientBytes: 19600 (Default)
Ports:
22
DCE/RPC 2 Preprocessor Configuration
Global Configuration
DCE/RPC Defragmentation: Enabled
Memcap: 102400 KB
Events: co
SMB Fingerprint policy: Disabled
Server Default Configuration
Policy: WinXP
Detect ports (PAF)
SMB: 139 445
TCP: 135
UDP: 135
RPC over HTTP server: 593
RPC over HTTP proxy: None
Autodetect ports (PAF)
SMB: None
TCP: 1025-65535
UDP: 1025-65535
RPC over HTTP server: 1025-65535
RPC over HTTP proxy: None
Invalid SMB shares: C$ D$ ADMIN$
Maximum SMB command chaining: 3 commands
SMB file inspection: Disabled
DNS config:
DNS Client rdata txt Overflow Alert: ACTIVE
Obsolete DNS RR Types Alert: INACTIVE
Experimental DNS RR Types Alert: INACTIVE
```

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```
rani@LAPTOP-OMML3RLS:/n  +  v
1 byte states : 1.02
2 byte states : 13.96
4 byte states : 0.00
-----
[ Number of patterns truncated to 20 bytes: 1038 ]
MaxRss at the end of detection rules:104752

--- Initialization Complete ---

--> Snort! <*-
Version 2.9.20 GRE (Build 82)
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using libpcap version 1.10.4 (with TPACKET_V3)
Using PCRE version: 8.39 2016-06-14
Using ZLIB version: 1.3

Rules Engine: SF_SNORT_DETECTION_ENGINE Version 3.2 <Build 1>
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>
Preprocessor Object: SF_5TCOMPLUS Version 1.0 <Build 1>
Preprocessor Object: SF_DCERPC2 Version 1.0 <Build 3>
Preprocessor Object: SF_DNS Version 1.1 <Build 4>
Preprocessor Object: SF_IMAP Version 1.0 <Build 1>
Preprocessor Object: SF_SIP Version 1.1 <Build 1>
Preprocessor Object: SF_POP Version 1.0 <Build 1>
Preprocessor Object: SF_SMTP Version 1.1 <Build 9>
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>
Preprocessor Object: SF_DNP3 Version 1.1 <Build 1>
Preprocessor Object: SF_SDF Version 1.1 <Build 1>
Preprocessor Object: SF_SSH Version 1.1 <Build 3>
Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>
Preprocessor Object: SF_GTP Version 1.1 <Build 1>
Preprocessor Object: SF_MODBUS Version 1.1 <Build 1>
Preprocessor Object: appid Version 1.1 <Build 5>

Total snort Fixed Memory Cost - MaxRss:104752
Snort successfully validated the configuration!
Snort exiting
rani@LAPTOP-OMML3RLS:/mnt/c/Users/Rani$ |
```

Step 5: Start Snort in Network Monitoring Mode

sudo snort -i eth0 -A console -c /etc/snort/snort.conf

```
rani@LAPTOP-OMML3RLS:/n  +  v
Snort successfully validated the configuration!
Snort exiting
rani@LAPTOP-OMML3RLS:/mnt/c/Users/Rani$ sudo snort -i eth0 -A console -c /etc/snort/snort.conf
Running in IDS mode

--- Initializing Snort ---
Initializing Output Plugins!
Initializing Preprocessors!
Initializing Plug-ins!
Parsing Rules file "/etc/snort/snort.conf"
PortVar 'HTTP_PORTS' defined : [ 80:81 311 383 591 593 901 1220 1414 1741 1830 2301 2381 2809 3037 3128 3702 4343 4848 5250 6988 7000:7001 7144:7145 7510 7
777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8888 8899 9000 9060 9080 9090:9091 9443 9999 11371 34444:34444 4108
0 50002 55555 ]
PortVar 'SHELLCODE_PORTS' defined : [ 0:79 81:65535 ]
PortVar 'ORACLE_PORTS' defined : [ 1824:65535 ]
PortVar 'SSH_PORTS' defined : [ 22 ]
PortVar 'FTP_PORTS' defined : [ 21 2100 3535 ]
PortVar 'SIP_PORTS' defined : [ 5060:5061 5600 ]
PortVar 'FILE_DATA_PORTS' defined : [ 80:81 110 143 311 383 591 593 901 1220 1414 1741 1830 2301 2381 2809 3037 3128 3702 4343 4848 5250 6988 7000:7001 714
4:7145 7510 7777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8888 8899 9000 9060 9080 9090:9091 9443 9999 11371 344
43:34444 41080 50002 55555 ]
PortVar 'GTP_PORTS' defined : [ 2123 2152 3386 ]
Detection:
  Search-Method = AC-Full-Q
  Split Any/Any group = enabled
  Search-Method-Optimizations = enabled
  Maximum pattern length = 20
Tagged Packet Limit: 256
Loading dynamic engine /usr/lib/snort/snort_dynamicengine/libsf_engine.so... done
Loading all dynamic detection libs from /usr/lib/snort/snort_dynamicrules...
WARNING: No dynamic libraries found in directory /usr/lib/snort/snort_dynamicrules.
  Finished loading all dynamic detection libs from /usr/lib/snort/snort_dynamicrules
Loading all dynamic preprocessor libs from /usr/lib/snort/snort_dynamicpreprocessor/...
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_appid_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_modbus_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_gtp_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_ftptelnet_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_ssh_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_sdf_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_dnp3_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_ssl_preproc.so... done
```

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```
rani@LAPTOP-OMML3RLS: /n
1 byte states : 1.02
2 byte states : 13.96
4 byte states : 0.00

-----
[ Number of patterns truncated to 20 bytes: 1038 ]
pcap DAQ configured to passive.
Acquiring network traffic from "eth0".
Reload thread starting...
Reload thread started, thread 0x77c6ca8dd6c0 (1925)
Decoding Ethernet

--- Initialization Complete ---

--> Snort! <--
o" )~
----
Version 2.0.20 GRE (Build 82)
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using libpcap version 1.10.4 (with TPACKET_V3)
Using PCRE version: 8.39 2016-06-14
Using ZLIB version: 1.3

Rules Engine: SF_SNORT_DETECTION_ENGINE Version 3.2 <Build 1>
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>
Preprocessor Object: SF_S7COMPLUS Version 1.0 <Build 1>
Preprocessor Object: SF_DCERPC2 Version 1.0 <Build 3>
Preprocessor Object: SF_DNS Version 1.1 <Build 4>
Preprocessor Object: SF_IMAP Version 1.0 <Build 1>
Preprocessor Object: SF_SIP Version 1.1 <Build 1>
Preprocessor Object: SF_POP Version 1.0 <Build 1>
Preprocessor Object: SF_SMTP Version 1.1 <Build 9>
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>
Preprocessor Object: SF_DNP3 Version 1.1 <Build 1>
Preprocessor Object: SF_SDF Version 1.1 <Build 1>
Preprocessor Object: SF_SSH Version 1.1 <Build 3>
Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>
Preprocessor Object: SF_GTP Version 1.1 <Build 1>
Preprocessor Object: SF_MODBUS Version 1.1 <Build 1>
Preprocessor Object: appid Version 1.1 <Build 5>

Commencing packet processing (pid=1924)
```

Step 6: Verify the Local Rules Inclusion

`sudo nano /etc/snort/rules/local.rules`

```
rani@LAPTOP-OMML3RLS: /n
GNU nano 7.2 /etc/snort/rules/local.rules
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
#
# LOCAL RULES
#
# This file intentionally does not come with signatures. Put your local
# additions here.
```

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```
rani@LAPTOP-OMML3RSL:/n  X + v
Total Statistics:
  Memory in use:      1379 bytes
  No of allocs:       3
  No of frees:        48
Config Statistics:
  Memory in use:      1379 bytes
  No of allocs:       3
  No of frees:        48
=====
Memory Statistics for File at:Wed Aug 26 09:49:45 2026
Total buffers allocated:      0
Total buffers freed:          0
Total buffers released:       0
Total file mempool:           0
Total allocated file mempool: 0
Total freed file mempool:     0
Total released file mempool:   0
Heap Statistics of file:
  Total Statistics:
    Memory in use:      280 bytes
    No of allocs:       6
    No of frees:        1
  Session Statistics:
    Memory in use:      0 bytes
    No of allocs:       1
    No of frees:        1
  Mempool Statistics:
    Memory in use:      280 bytes
    No of allocs:       5
    No of frees:        0
=====
Snort exiting
rani@LAPTOP-OMML3RSL:/mnt/c/Users/Rani$ sudo nano /etc/snort/rules/local.rules
rani@LAPTOP-OMML3RSL:/mnt/c/Users/Rani$ grep "local.rules" /etc/snort/snort.conf
grep: /etc/snort/snort.conf: Permission denied
rani@LAPTOP-OMML3RSL:/mnt/c/Users/Rani$ sudo grep "local.rules" /etc/snort/snort.conf
include $RULE_PATH/local.rules
rani@LAPTOP-OMML3RSL:/mnt/c/Users/Rani$ |
```

Step 7: Test the Snort Configuration

```
rani@LAPTOP-OMML3RSL:/n  X + v
include $RULE_PATH/local.rules
rani@LAPTOP-OMML3RSL:/mnt/c/Users/Rani$ sudo snort -T -c /etc/snort/snort.conf
Running in Test mode

--- Initializing Snort ---
Initializing Output Plugins!
Initializing Preprocessors!
Initializing Plug-ins!
Parsing Rules file "/etc/snort/snort.conf"
PortVar 'HTTP_PORTS' defined : [ 80:81 311 383 591 593 901 1220 1414 1741 1830 2301 2381 2809 3037 3128 3702 4343 4848 5250 6988 7000:7001 7144:7145 7510 7
777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8888 8899 9000 9060 9080 9090:9091 9443 9999 11371 34443:34444 4108
0 50002 55555 ]
PortVar 'SHELLCODE_PORTS' defined : [ 0:79 81:65535 ]
PortVar 'ORACLE_PORTS' defined : [ 1024:65535 ]
PortVar 'SSH_PORTS' defined : [ 22 ]
PortVar 'FTP_PORTS' defined : [ 21 2100 3535 ]
PortVar 'SIP_PORTS' defined : [ 5060:5061 5600 ]
PortVar 'FILE_DATA_PORTS' defined : [ 80:81 110 143 311 383 591 593 901 1220 1414 1741 1830 2301 2381 2809 3037 3128 3702 4343 4848 5250 6988 7000:7001 714
4:7145 7510 7777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8888 8899 9000 9060 9080 9090:9091 9443 9999 11371 344
43:34444 41080 50002 55555 ]
PortVar 'GTP_PORTS' defined : [ 2123 2152 3386 ]
Detection:
  Search-Method = AC-Full-Q
  Split Any/Any group = enabled
  Search-Method-Optimizations = enabled
  Maximum pattern length = 20
Tagged Packet Limit: 256
Loading dynamic engine /usr/lib/snort/snort_dynamicengine/libsengine.so... done
Loading all dynamic detection libs from /usr/lib/snort/snort_dynamicrules...
WARNING: No dynamic libraries found in directory /usr/lib/snort/snort_dynamicrules.
Finished Loading all dynamic detection libs from /usr/lib/snort/snort_dynamicrules
Loading all dynamic preprocessor libs from /usr/lib/snort/snort_dynamicpreprocessor/...
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_appid_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_modbus_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_gtp_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_ftptelnet_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_ssh_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_sdf_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_dnp3_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_ssl_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_sntp_preproc.so... done
```

SHETH L.U.J AND SIR MV COLLEGE PRACTICAL NO. 08

Step 8: Start Snort IDS for ICMP Detection

sudo snort -i eth0 -A console -c /etc/snort/snort.conf

```
rani@LAPTOP-OMML3RLS: /n  x  +  v
2 byte states : 13.96
4 byte states : 0.00
-----
[ Number of patterns truncated to 20 bytes: 1038 ]
pcap DAQ configured to passive.
Acquiring network traffic from "eth0".
Reload thread starting...
Reload thread started, thread 0x77aafd4fd6c0 (2027)
Decoding Ethernet

--== Initialization Complete ==--

--> Snort! <*-
o" )~
----
Version 2.9.20 GRE (Build 82)
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using libpcap version 1.10.4 (with TPACKET_V3)
Using PCRE version: 8.39 2016-06-14
Using ZLIB version: 1.3

Rules Engine: SF_SNORT_DETECTION_ENGINE Version 3.2 <Build 1>
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>
Preprocessor Object: SF_S7COMPLUS Version 1.0 <Build 1>
Preprocessor Object: SF_DCEP2C2 Version 1.0 <Build 3>
Preprocessor Object: SF_DNS Version 1.1 <Build 4>
Preprocessor Object: SF_IMAP Version 1.0 <Build 1>
Preprocessor Object: SF_SIP Version 1.1 <Build 1>
Preprocessor Object: SF_POP Version 1.0 <Build 1>
Preprocessor Object: SF_SMTP Version 1.1 <Build 9>
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>
Preprocessor Object: SF_DNP3 Version 1.1 <Build 1>
Preprocessor Object: SF_SDF Version 1.1 <Build 1>
Preprocessor Object: SF_SSH Version 1.1 <Build 3>
Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>
Preprocessor Object: SF_GTP Version 1.1 <Build 1>
Preprocessor Object: SF_MODBUS Version 1.1 <Build 1>
Preprocessor Object: appid Version 1.1 <Build 5>
Commencing packet processing (pid=2026)
08/26-10:00:59.285355  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {IPV6-ICMP} fe80::215:5dff:fe3b:c325 -> ff02::2
```

Step 10: Stop Snort

```
rani@LAPTOP-OMML3RLS: /n  x  +  v
No of Allocs : 3
No of Frees : 48
Total memory used : 1379

Heap Statistics of imap:
Total Statistics:
Memory in use: 1379 bytes
No of allocs: 3
No of frees: 48
Config Statistics:
Memory in use: 1379 bytes
No of allocs: 3
No of frees: 48
=====
Memory Statistics for File at:Wed Aug 26 10:02:31 2026
Total buffers allocated: 0
Total buffers freed: 0
Total buffers released: 0
Total file mempool: 0
Total allocated file mempool: 0
Total freed file mempool: 0
Total released file mempool: 0

Heap Statistics of file:
Total Statistics:
Memory in use: 280 bytes
No of allocs: 6
No of frees: 1
Session Statistics:
Memory in use: 0 bytes
No of allocs: 1
No of frees: 1
Mempool Statistics:
Memory in use: 280 bytes
No of allocs: 5
No of frees: 0
=====
Snort exiting
rani@LAPTOP-OMML3RLS: /mnt/c/Users/Rani$
```

SHETH L.U.J AND SIR MV COLLEGE

PRACTICAL NO. 08

Generate Test Traffic (ping 172.23.187.84)

```
rani@LAPTOP-OMML3RLS: /m  x  rani@LAPTOP-OMML3RLS: /m  x  +  v
64 bytes from 172.23.187.84: icmp_seq=152 ttl=64 time=0.153 ms
64 bytes from 172.23.187.84: icmp_seq=153 ttl=64 time=0.070 ms
64 bytes from 172.23.187.84: icmp_seq=154 ttl=64 time=0.058 ms
64 bytes from 172.23.187.84: icmp_seq=155 ttl=64 time=0.078 ms
64 bytes from 172.23.187.84: icmp_seq=156 ttl=64 time=0.153 ms
64 bytes from 172.23.187.84: icmp_seq=157 ttl=64 time=0.085 ms
64 bytes from 172.23.187.84: icmp_seq=158 ttl=64 time=0.092 ms
64 bytes from 172.23.187.84: icmp_seq=159 ttl=64 time=0.098 ms
64 bytes from 172.23.187.84: icmp_seq=160 ttl=64 time=0.080 ms
64 bytes from 172.23.187.84: icmp_seq=161 ttl=64 time=0.084 ms
64 bytes from 172.23.187.84: icmp_seq=162 ttl=64 time=0.115 ms
64 bytes from 172.23.187.84: icmp_seq=163 ttl=64 time=0.083 ms
64 bytes from 172.23.187.84: icmp_seq=164 ttl=64 time=0.090 ms
64 bytes from 172.23.187.84: icmp_seq=165 ttl=64 time=0.076 ms
64 bytes from 172.23.187.84: icmp_seq=166 ttl=64 time=0.075 ms
64 bytes from 172.23.187.84: icmp_seq=167 ttl=64 time=0.087 ms
64 bytes from 172.23.187.84: icmp_seq=168 ttl=64 time=0.086 ms
64 bytes from 172.23.187.84: icmp_seq=169 ttl=64 time=0.089 ms
64 bytes from 172.23.187.84: icmp_seq=170 ttl=64 time=0.134 ms
64 bytes from 172.23.187.84: icmp_seq=171 ttl=64 time=0.108 ms
64 bytes from 172.23.187.84: icmp_seq=172 ttl=64 time=0.089 ms
64 bytes from 172.23.187.84: icmp_seq=173 ttl=64 time=0.092 ms
64 bytes from 172.23.187.84: icmp_seq=174 ttl=64 time=0.097 ms
64 bytes from 172.23.187.84: icmp_seq=175 ttl=64 time=0.097 ms
64 bytes from 172.23.187.84: icmp_seq=176 ttl=64 time=0.153 ms
64 bytes from 172.23.187.84: icmp_seq=177 ttl=64 time=0.095 ms
64 bytes from 172.23.187.84: icmp_seq=178 ttl=64 time=0.088 ms
64 bytes from 172.23.187.84: icmp_seq=179 ttl=64 time=0.070 ms
64 bytes from 172.23.187.84: icmp_seq=180 ttl=64 time=0.099 ms
64 bytes from 172.23.187.84: icmp_seq=181 ttl=64 time=0.073 ms
64 bytes from 172.23.187.84: icmp_seq=182 ttl=64 time=0.073 ms
64 bytes from 172.23.187.84: icmp_seq=183 ttl=64 time=0.102 ms
64 bytes from 172.23.187.84: icmp_seq=184 ttl=64 time=0.084 ms
64 bytes from 172.23.187.84: icmp_seq=185 ttl=64 time=0.110 ms
64 bytes from 172.23.187.84: icmp_seq=186 ttl=64 time=0.076 ms
64 bytes from 172.23.187.84: icmp_seq=187 ttl=64 time=0.068 ms
^C
--- 172.23.187.84 ping statistics ---
187 packets transmitted, 187 received, 0% packet loss, time 185904ms
rtt min/avg/max/mdev = 0.057/0.298/34.821/2.539 ms
rani@LAPTOP-OMML3RLS: /mnt/c/Users/Rani$
```

```
rani@LAPTOP-OMML3RLS: /m  x  rani@LAPTOP-OMML3RLS: /m  x  +  v
No of Allocs :      3
No of Frees  :     48
Total memory used : 1379

Heap Statistics of imap:
  Total Statistics:
    Memory in use: 1379 bytes
    No of allocs:  3
    No of frees:  48
  Config Statistics:
    Memory in use: 1379 bytes
    No of allocs:  3
    No of frees:  48
=====

Memory Statistics for File at:Wed Aug 26 10:13:22 2026

Total buffers allocated:      0
Total buffers freed:         0
Total buffers released:      0
Total file mempool:          0
Total allocated file mempool: 0
Total freed file mempool:    0
Total released file mempool:  0

Heap Statistics of file:
  Total Statistics:
    Memory in use: 280 bytes
    No of allocs:  6
    No of frees:  1
  Session Statistics:
    Memory in use:  0 bytes
    No of allocs:  1
    No of frees:  1
  Mempool Statistics:
    Memory in use: 280 bytes
    No of allocs:  5
    No of frees:  0
=====

Snort exiting
rani@LAPTOP-OMML3RLS: /mnt/c/Users/Rani$
```